Restaurant Management System

“Database Management Project”

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Course: Database Management System

Introduction:

The restaurant management system is a digital solution designed to streamline daily operations in the food service industry. By automating key processes, this system replaces manual record-keeping with a reliable database environment. The primary objective is to manage the restaurant's core components—specifically employees, menu items, and customer orders—in a centralized and efficient manner. This project highlights the practical application of database design principles to ensure data integrity, faster service, and accurate tracking of restaurant activities.

Problem Statement:

Managing restaurant operations manually leads to several challenges, including data loss, potential human errors in recording orders, and difficulty in tracking staff activities. Furthermore, retrieving historical order data and maintaining an accurate inventory of menu items become cumbersome tasks. This project addresses these issues by implementing a structured database management system, which centralizes information and ensures accuracy, reliability, and efficiency in all restaurant-related transactions.

Methodology:

The development of this restaurant management system followed a structured approach to ensure optimal database performance:

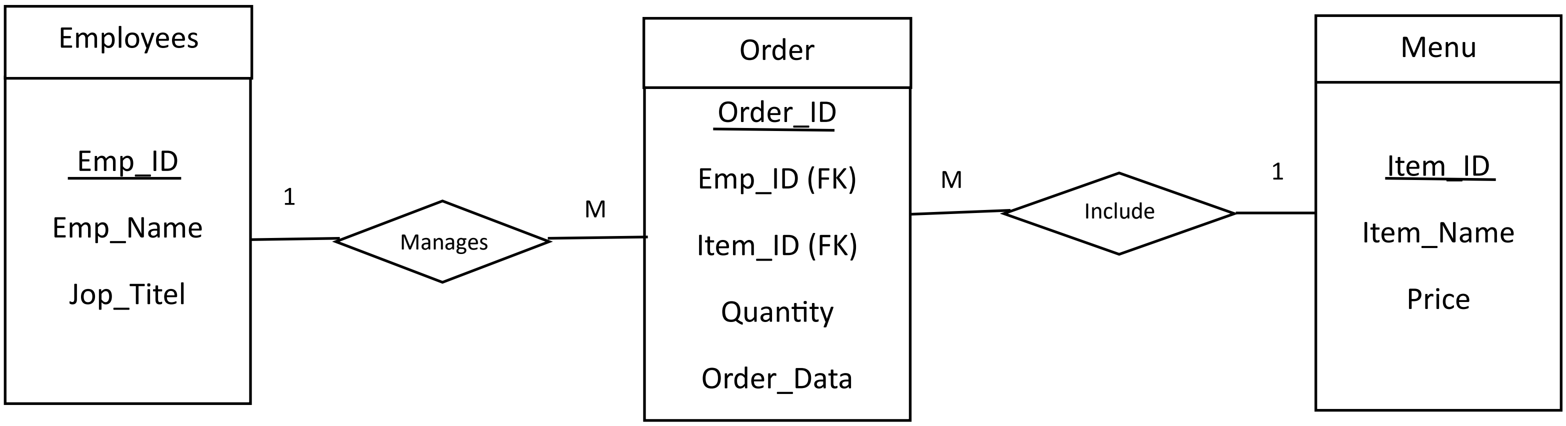
*Requirements Analysis: Identifying the key entities (Employees, Orders, Menu) and their relationships.

*Conceptual Design: Drawing the Entity-Relationship Diagram (ERD) to visualize the database structure.

*Logical Implementation: Mapping the ERD into relational tables using SQL to define primary and foreign keys.

*Testing: Ensuring that data insertion, updates, and queries function correctly without errors

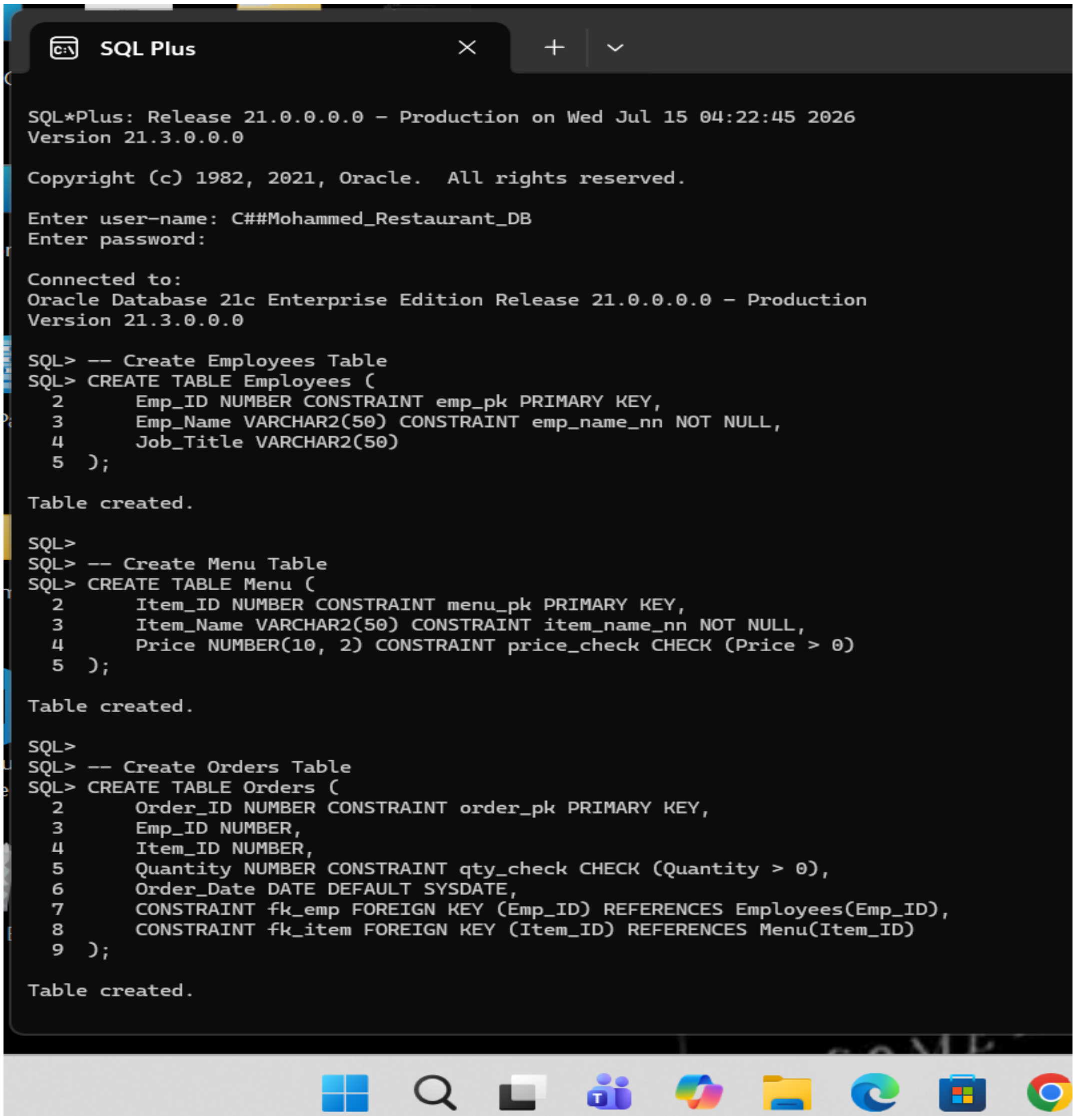
Database Design (ER Diagram)



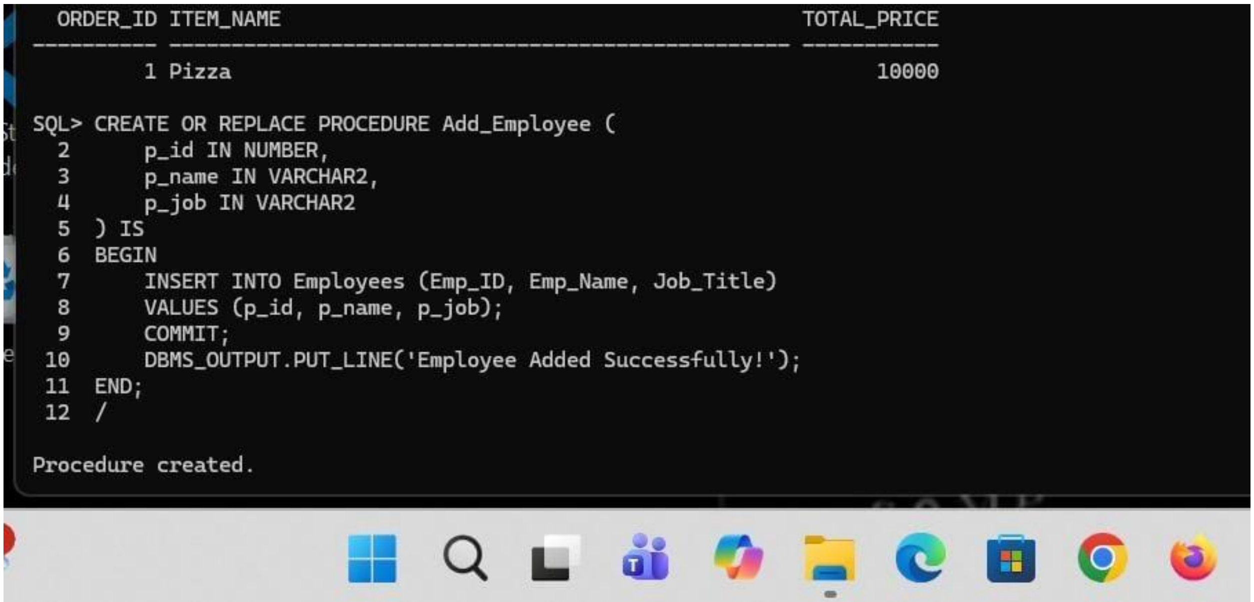
Description:

The Entity-Relationship Diagram (ERD) illustrates the logical structure of the database. It defines three main entities: Employees, Orders, and Menu.

SQL Implementation:



This screenshot shows the creation of the database tables (Employees, Menu, and Orders) using Oracle SQL*Plus. The code defines the structure, primary keys, and relationships between these tables to ensure the database works correctly.



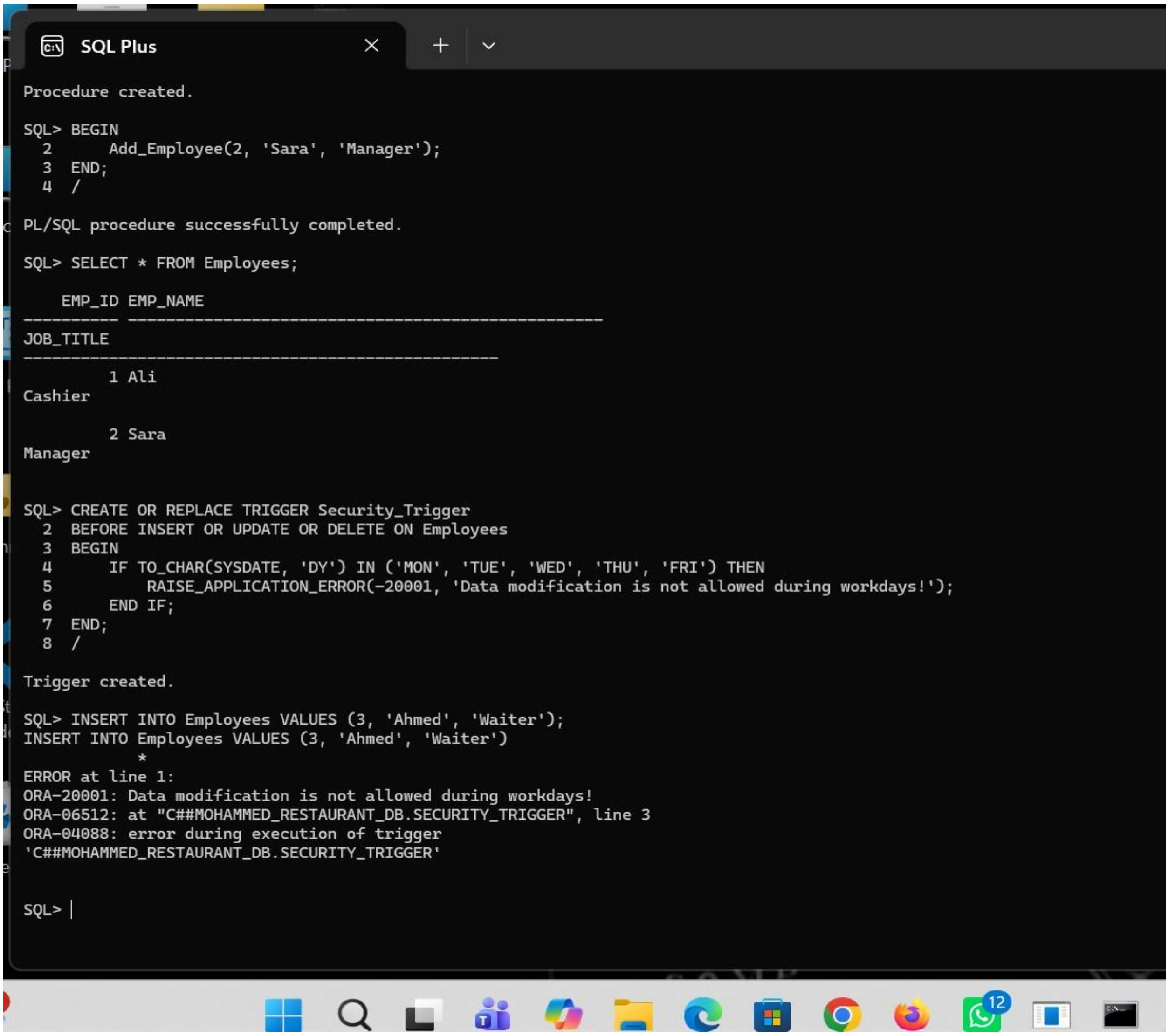
```
  ORDER_ID ITEM_NAME                                          TOTAL_PRICE
---------- -------------------------------------------------- -----------
         1 Pizza                                                    10000

SQL> CREATE OR REPLACE PROCEDURE Add_Employee (
  2      p_id IN NUMBER,
  3      p_name IN VARCHAR2,
  4      p_job IN VARCHAR2
  5  ) IS
  6  BEGIN
  7      INSERT INTO Employees (Emp_ID, Emp_Name, Job_Title)
  8      VALUES (p_id, p_name, p_job);
  9      COMMIT;
 10      DBMS_OUTPUT.PUT_LINE('Employee Added Successfully!');
 11  END;
 12  /

Procedure created.
```

Implementation Note:

The Add_Order procedure was created to automate the process of inserting new records into the Orders table. This enhances data integrity and simplifies the transaction process by automatically setting the Order_Date to the current system date.



```
SQL Plus

Procedure created.

SQL> BEGIN
  2      Add_Employee(2, 'Sara', 'Manager');
  3  END;
  4  /

PL/SQL procedure successfully completed.

SQL> SELECT * FROM Employees;

    EMP_ID EMP_NAME
---------- --------------------------------------------------
JOB_TITLE
--------------------------------------------------
         1 Ali
Cashier

         2 Sara
Manager


SQL> CREATE OR REPLACE TRIGGER Security_Trigger
  2  BEFORE INSERT OR UPDATE OR DELETE ON Employees
  3  BEGIN
  4      IF TO_CHAR(SYSDATE, 'DY') IN ('MON', 'TUE', 'WED', 'THU', 'FRI') THEN
  5          RAISE_APPLICATION_ERROR(-20001, 'Data modification is not allowed during workdays!');
  6      END IF;
  7  END;
  8  /

Trigger created.

SQL> INSERT INTO Employees VALUES (3, 'Ahmed', 'Waiter');
INSERT INTO Employees VALUES (3, 'Ahmed', 'Waiter')
            *
ERROR at line 1:
ORA-20001: Data modification is not allowed during workdays!
ORA-06512: at "C##MOHAMMED_RESTAURANT_DB.SECURITY_TRIGGER", line 3
ORA-04088: error during execution of trigger
'C##MOHAMMED_RESTAURANT_DB.SECURITY_TRIGGER'


SQL> |
```

Trigger Implementation:

The Security_Trigger was created to protect data integrity by blocking unauthorized modifications to the Employees table during workdays.

Results:

*The database system successfully maintains data consistency and integrity through primary and foreign key constraints.

*Stored procedures and triggers have been implemented to automate order processing and enforce security rules.

*System testing confirms that all modules function correctly, effectively preventing unauthorized operations and ensuring efficient data management.

Conclusion:

This project successfully designed and implemented a functional restaurant database system using Oracle SQL. By integrating entity-relationship modeling, automated procedures, and security triggers, the system ensures efficient data management and protection. Ultimately, this implementation provides a robust and reliable foundation for managing restaurant operations, demonstrating the practical application of database theory in a real-world business context.